

CO3-AT3.

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COURSE CODE: ITAO304

1.

Given : Data packet Size = 500 bytes

Tunnel overhead = 20 bytes per packet

Total Size = Original data + Overhead

$$= 500 + 20$$

$$= 520 \text{ bytes}$$

$$\text{Overhead \%} = \frac{20}{500} \times 100$$

$$= 4\%$$

2.

Given : Bandwidth = 20 MHz

Modulation = 64 QAM

Each symbol carries = 6 bits.

Step 1: Convert bandwidth $20 \text{ MHz} = 20 \times 10^6 \text{ Hz}$

$$\text{Symbol rate} = 20 \times 10^6$$

Step 2: Data Rate = Symbol Rate $\times$ Bits per

$$= 20 \times 10^6 \times 6$$

$$= 120 \times 10^6 \text{ bits/sec}$$

$$\therefore 120 \times 10^6 = 120 \text{ Mbps}$$

QAM^{Quadrature} is a digital modulation technique

used in wireless communication to transmit data by changing amplitude and phase.

3,

Given:

No. of routing table entries = 50
 Memory required for each entry = 32 bytes.

$$\begin{aligned}\text{Total memory} &= \text{No. of entries} \times \text{each entry memory} \\ &= 50 \times 32 \\ &= 1600 \text{ bytes.}\end{aligned}$$

Convert to KB: $1600 \div 1024 \approx 1.56 \text{ KB.}$

Given:

Speed of mobile node = 10 m/s

$$\text{Distance} = \text{Speed} \times \text{Time}$$

$$= 10 \times 2$$

$$= 20 \text{ m.}$$

Given:

Intermediate nodes = 5.

Total 5 RREQ forwarding transmissions.

$$\Rightarrow \text{Total} = 1 + 5 = 6.$$

$$\therefore \text{Total Route Request packets} = 6.$$

Given:

$$\text{Subnet mask} = 255.255.255.240.$$

$$240 = 11110000$$

$$\therefore 32 - 28 = 4 \text{ host bits.}$$

Find total IP addresses

$$= 2^4 = 16.$$

$\therefore$ There are 16 total IP addresses.

1 network address, 1 broadcast address

$$= 16 - 2 = 14.$$

$\therefore$ 14 mobile nodes.

Given:

$$\text{channel bandwidth} = 20 \text{ MHz.}$$

$$\text{Symbol rate} = 1 \text{ Msps.}$$

$$\text{frequency} = 2.4 \text{ GHz}$$

$$20 \text{ MHz} = 20 \times 10^6 \text{ Hz.}$$

$$1 \text{ Msps} = 1 \times 10^6 \text{ symbols/sec}$$

$$\text{Spectral Efficiency} = \frac{\text{Data Rate}}{\text{Bandwidth}}$$

$$\text{Data Rate} = 1 \times 10^6 \text{ bps} = 1 \text{ Mbps.}$$

$$= \frac{1 \text{ Mbps}}{20 \text{ MHz}} = 0.05 \text{ bits/s/Hz.}$$

8,

Given:

Number of nodes = 50

Area = 1 km².

$$\text{Density} = \frac{\text{No. of nodes}}{\text{Area}}$$

$$= \frac{50}{1} = 50 \text{ nodes / km}^2$$

Given:

2.50 m

 πr^2

62,500

9.82 nodes / km²

9,

Given:mobile node speed $v = 30 \text{ m/s}$ frequency, $f = 900 \text{ MHz}$.Speed of light, $c = 3 \times 10^8 \text{ m/s}$.

Doppler Shift: Is the change in frequency of a wave when the source and observer are moving relative to each other.

$$f_d = \frac{vf}{c}$$

$$f_d = \frac{30 \times 900 \times 10^6}{3 \times 10^8}$$

$$= 90,000 \text{ Hz}$$

$$= 90 \text{ kHz}$$

$$= \boxed{f_d = 90 \text{ kHz}}$$

10,

Given:

Data rate = 54 Mbps

packet size = 1500 bytes.

1 byte = 8 bits.

$$\text{Size} = 1500 \times 8 = 12000 \text{ bits}$$

$$54 \text{ Mbps} = 54 \times 10^6 \text{ bits/sec}$$

$$\text{Transmission Time} = \frac{12000}{54 \times 10^6} = 0.0002222 \text{ seconds}$$

convert to milliseconds

$$: 0.0002222 \times 1000 = 0.2222 \text{ ms}$$